2023 AMC 10A Problem 9 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 9

Problem:

A digital display shows the current date as an 8-digit integer consisting of a 4-digit year, followed by a 2-digit month, followed by a 2-digit date within the month. For example, Arbor Day this year is displayed as 20230428. For how many dates in 2023 does each digit appear an even number of times in the 8-digit display for that date?

Answer Choices:

A. 5

B. 6

C. 7

D. 8

E. 9

Solution:

Because the first four digits of the display must be 2023, the last four digits must contain one 0 , one 3 , and two of the same digit. These digits cannot contain three 0 s or three 3 s , so the only possible repeated digits among them are 1 and 2.

If the repeated digit is 1 , the possible dates are $01/13, 01/31, 03/11, 10/13, 10/31, 11/03$, and $11/30$. If the repeated digit is 2 , the possible dates are $02/23$ and $03/22$. In all there are $\text{(E)}\boxed{9}$ dates in 2023 for which each digit appear an even number of times.

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